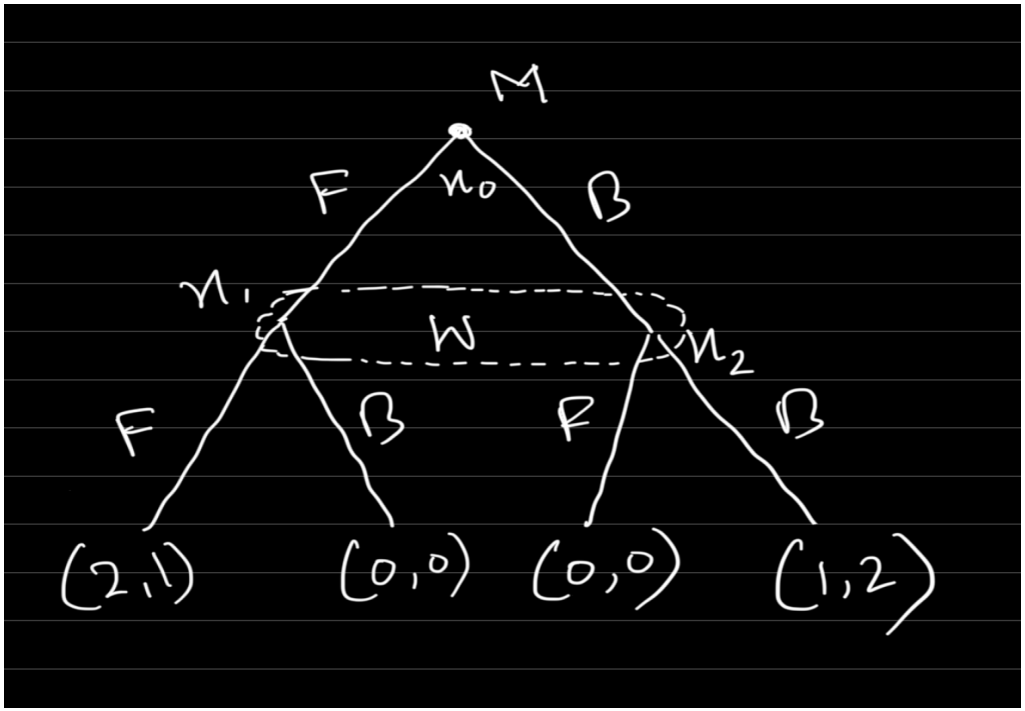
Extensive-form games

Extensive form games with imperfect information

- Recall the idea of perfect information–whenever a player is supposed to take an action, the player knows precisely the node where the game is currently at.
- This does not hold for imperfect information games.

Example

- Consider the original battle of the sexes game where the man and the woman simultaneously choose their actions.
- The game can be represented in the extensive form in the following way.



Example

- Here a new element to the depictions of extensive-form games: the two nodes x_1 and x_2 of the woman are surrounded by an ellipse.
- This represents the fact that when the woman is in the position of selecting between F and B, she does not know whether the game state is currently at node x_1 or node x_2, because she does not know whether the man has selected F or B.
- These two nodes together form an **information set** of the woman.

Example

- Note that the game battle of the sexes is symmetric between the man and the woman, but the game tree shown in Figure 1 is not symmetric.
- The figure depicts the man making his choice before the woman's choice, with the woman not knowing which choice the man made; this is done in order to depict the game conveniently as a tree.
- Alternatively, we could have drawn the game tree with the woman making the first move, and then the man chooses his action without known what the woman has chosen.

Example

- Here is the alternative representation.

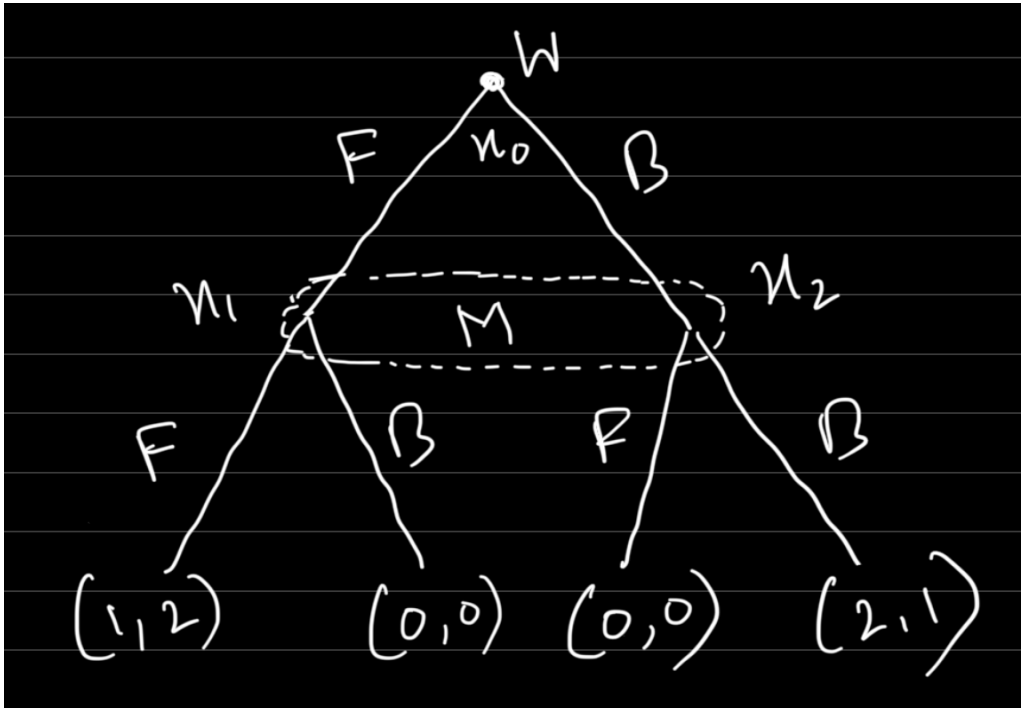


Figure: Figure 2

Example

- Consider another example as shown in the following figure, where it's imperfect information with chance moves.

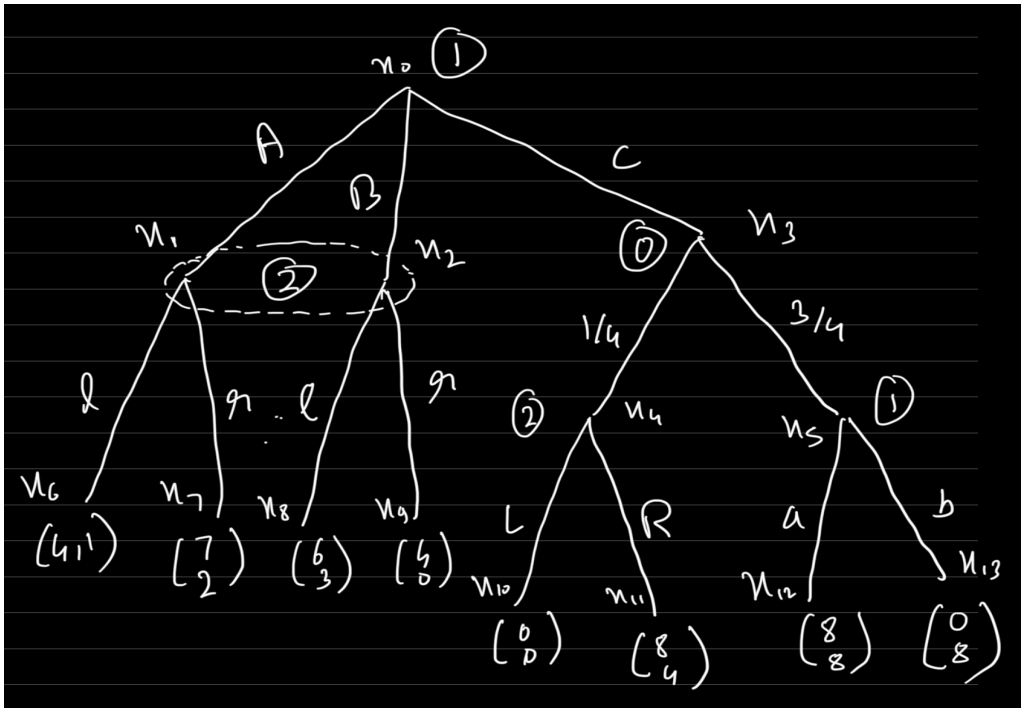


Figure: Figure 3

Example

- Here, only Player 2 has a non-trivial information set (containing more than one node), and there is a chance node.
- Note that the fact that Player 2 has the same set of actions at each of the two nodes in this information set is a necessary consequence.
- If this were not the case, Player 2 would know at which node he is (i.e., would know whether Player 1 had played A or B) by simply examining the set of available actions, which would go against the interpretation of an information set.

Formal definition

Definition

A game in extensive form (with imperfect information and chance moves) is a tuple

$$\Gamma = \langle N, V, E, x_0, (V_i)_{i \in N \cup \{0\}}, \mathcal{H} = (H_i)_{i \in N}, \mathcal{A} = (A(h))_{h \in \mathcal{H}}, (p_x)_{x \in V_0}, (u_i)_{i \in N} \rangle$$

where

- N is a finite set of players.
- (V, E, x_0) is a directed rooted tree, called the game tree.
- $(V_i)_{i \in N \cup \{0\}}$ is a partition of the set of vertices that are not leaves among the agents and nature (Player 0), i.e, $V \setminus Z$.

Formal definition

Definition

- $\mathcal{H} = (H_i)_{i \in N}$ where for each $i \in N$, H_i is a partition of the set V_i. The sets $h \in H_i$ are called information sets of player i. Each $h \in H_i$ is assumed to satisfy

  (i) every node in h has the same number of outgoing edges, and

  (ii) every path in (V, E, x_0) intersects h at most once.

Formal definition

Definition

- $\mathcal{A} = (A(h))_{h \in H}$, where $H = \cup_{i \in N} H_i$, and for each $h \in H$, $A(h)$ is a partition of the set of edges outgoing from nodes $x \in h$. The partition $A(h)$ is such that: for each $x \in h$ and each $a \in A(h)$, a contains exactly one edge outgoing from x.

  A set $a \in A(h)$ is called an action at h.

Formal definition

Definition

- For every vertex $x \in V_0$, p_x is a probability distribution over the edges outgoing from x.
- $(u_i)_{i \in N}$ is the collection of utility functions where $u_i : Z \to \mathbb{R}$ for all $i \in N$.

Example

- Consider the following example:

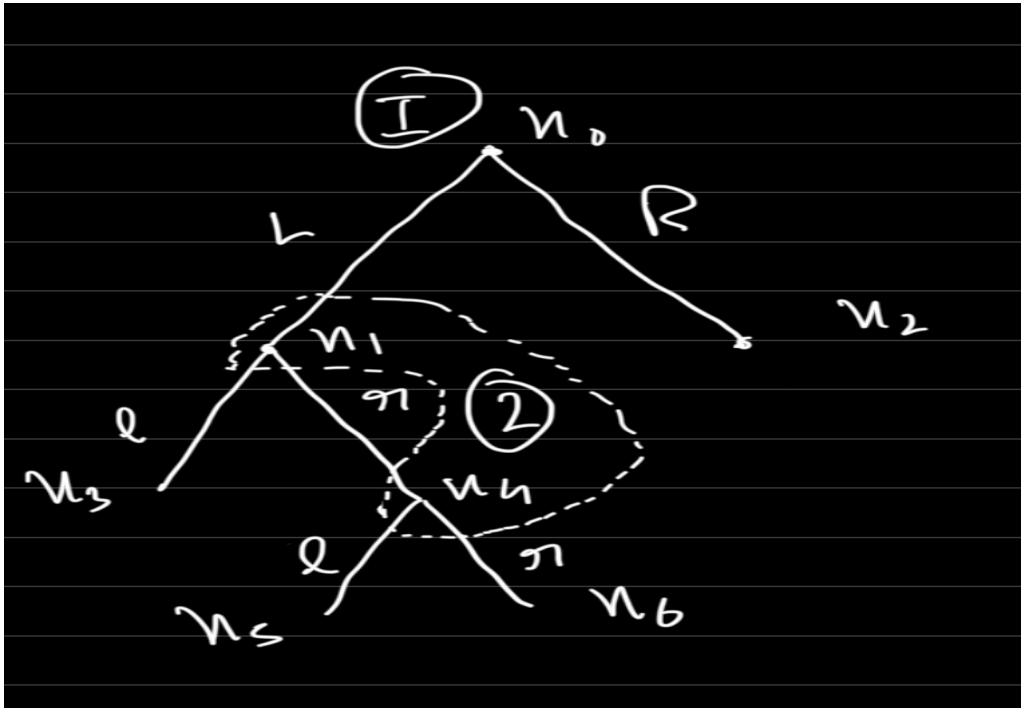


Figure: Figure 4

- Is this a valid extensive form game?

Example

- The answer is "no", as the path (x_0, x_1, x_4, x_6) intersects the information set of Player 2 twice, which is not allowed by the definition of an information set.

Example

- Consider the following example:

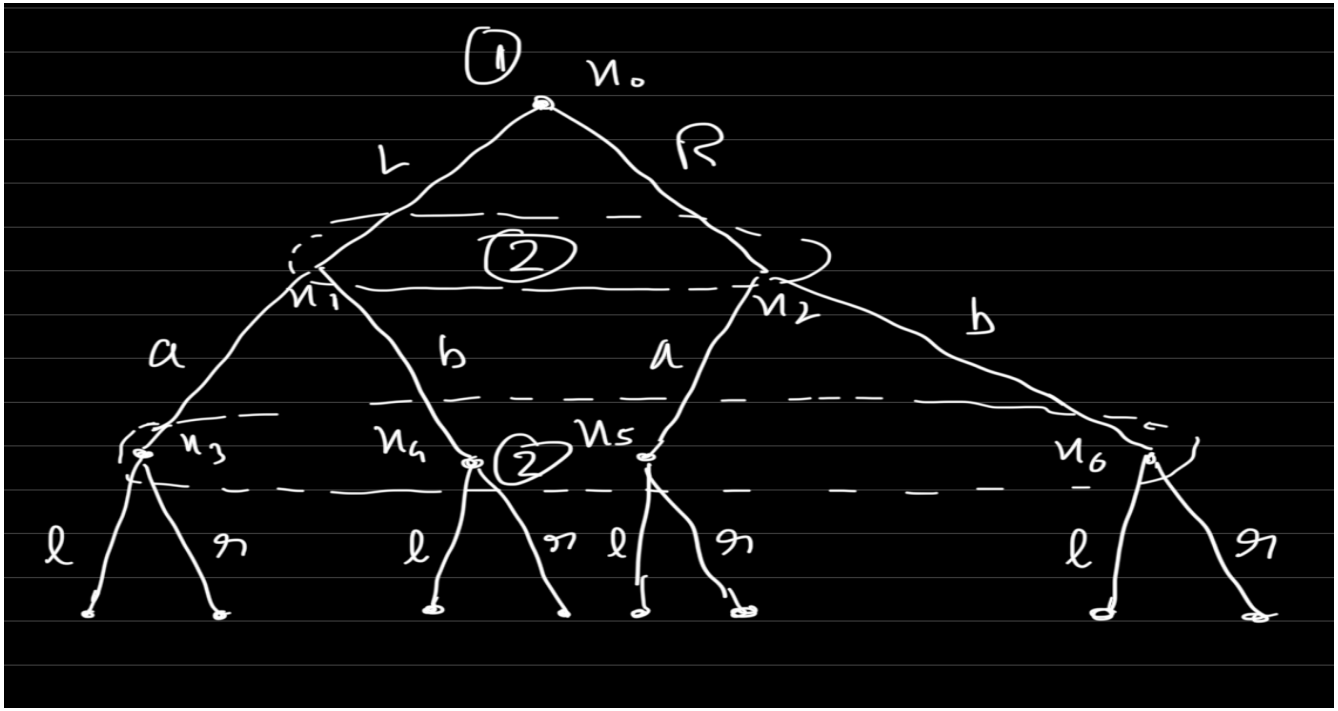


Figure: Figure 5

- Is this a valid extensive form game?

Example

- The answer is "yes". It satisfies all the properties of an extensive form game with imperfect information. However, here, Player 2 forgets what he did before. If you consider x_4 and x_5 in Player 2's second information set, the paths from the root to these nodes contain different actions of Player 2.
- We define the notion of perfect recall that says at any point of the game, the players remember the actions they have taken before.

Perfect recall

Definition

An extensive form game $\mathcal{S}$ has perfect recall for Player i if for all information set $h \in H_i$ and for all $x, y \in h$, an action of Player i is on the path from the root to x if and only if it is on the path from the to y.

A game satisfies perfect recall if it satisfies perfect recall for all the players.

Example

- Consider the following game.

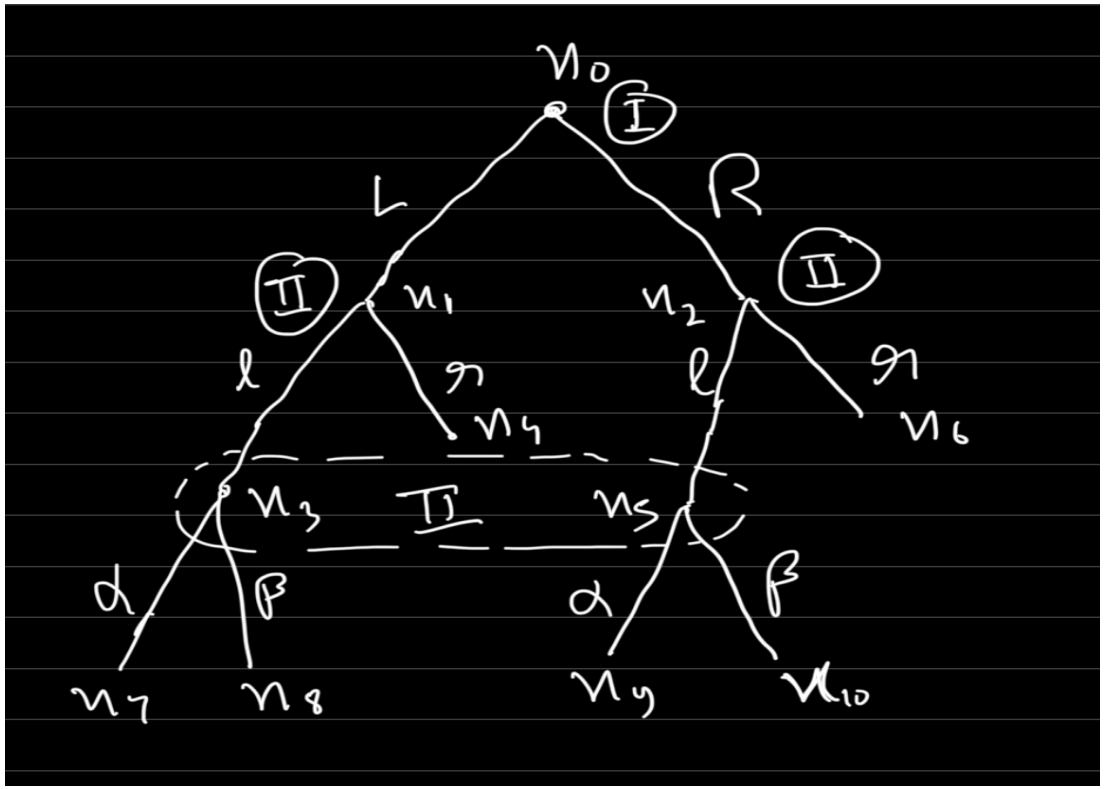


Figure: Figure 6

- Does the game have perfect recall for Player 2?

Example

- Note that this does not satisfy perfect recall for Player 2. Although both the actions are denoted by the same letter l but they are different in terms of what is known to Payer 2 while taking those actions.

Example

- Consider the following game.

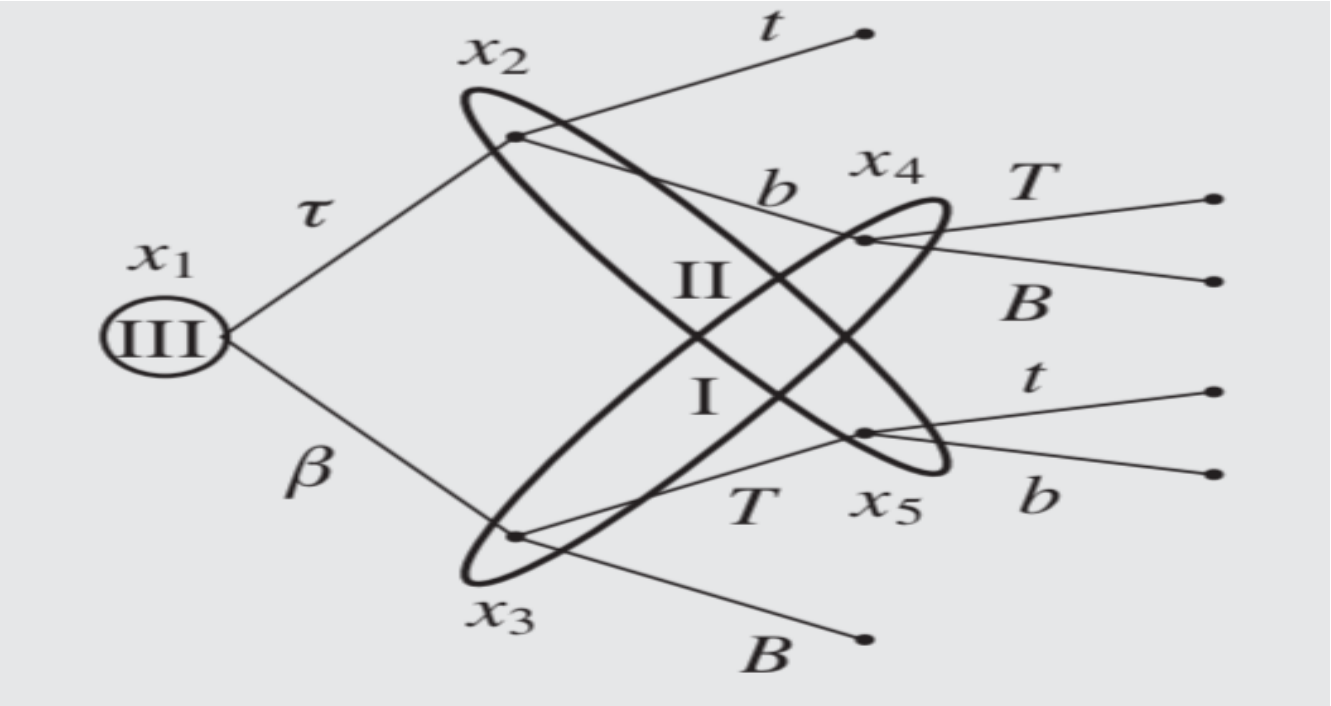


Figure: Figure 7

- Does the game have perfect recall for all the players?

Example

- Yes, verify that it satisfies the condition for all the players.